

Getting MST Testnet Funds (MSTC)

Step-by-step procedure for B1: create the 5 system wallets, claim testnet MSTC from the official faucet, distribute it, and verify every balance before deploying the contracts.

Network reference

Network	MST Testnet
Chain ID	91562037
RPC (HTTP)	https://testnetrpc.mstblockchain.com
RPC (WebSocket)	wss://testnetrpc.mstblockchain.com
Gas token	MSTC (18 decimals)
Faucet (official)	https://faucet.mstblockchain.com/
Explorer	https://testnet.mstscan.com (community-reported; confirm it loads)
EVM version	Cancun

Golden rule: private keys are never typed, pasted, printed or shown to any AI assistant. Keys live only in backend/.env (git-ignored). The faucet and every check below need the **address** only.

Step 1 — Install the tools

- **Foundry** (provides cast, used to create wallets, check balances and send funds).
- **MetaMask** browser extension (optional; for viewing balances visually).
- **Git** and the cloned repo, on your own branch.

```
curl -L https://foundry.paradigm.xyz | bash
foundryup
cast --version          # must print a version number
```

Step 2 — Make sure .env can never be committed

Run from the repo root. It must print a .gitignore rule. If it prints nothing, stop and add .env to .gitignore first.

```
git check-ignore -v backend/.env
```

Step 3 — Create the 5 system wallets

Run the wallet script (written by the B1 agent). It writes each key to backend/.env and prints **only addresses**. Run it yourself in your terminal, not through the AI.

```
bash backend/scripts/gen-wallets.sh

# expected output (addresses only):
# ORG          0x....
# AGENT        0x....
# ARBITRATOR   0x....
# DEMO_BUYER   0x....
# DEMO_SELLER  0x....
```

If the script does not exist yet, create one wallet at a time with `cast wallet new` **in your own terminal**, copy the key into backend/.env by hand using a text editor, and clear the terminal afterwards.

Copy the five **addresses** (never keys) into deployments.md under "System wallets".

Step 4 — Add MST Testnet to MetaMask (optional)

- MetaMask → network selector → **Add network** → **Add a network manually**.
- Enter: Network name MST Testnet, RPC URL <https://testnetrpc.mstblockchain.com>, Chain ID 91562037, Currency symbol MSTC, Explorer <https://testnet.mstscan.com> (optional).
- You do **not** need to import the system wallets into MetaMask. The faucet only needs an address.

Step 5 — Claim MSTC from the faucet into the ORG wallet

- Open <https://faucet.mstblockchain.com/> in a normal browser (the page needs JavaScript).
- Paste the **ORG** address from Step 3 and claim.
- Claim only into ORG. Faucets usually limit claims per address or per IP; ORG will fund the others in Step 7.

Step 6 — Verify the ORG balance

```
cast balance <ORG_ADDRESS> --rpc-url https://testnetrpc.mstblockchain.com --ether
```

A non-zero value (in MSTC) means the claim worked. With ~3-second blocks it should appear within seconds. If it stays at 0 after a minute, see Troubleshooting.

Step 7 — Distribute MSTC from ORG to the other 4 wallets

The key is loaded from `.env` into the shell; it is never typed or printed. 0.2ether here means 0.2 MSTC (both use 18 decimals). Adjust the amount to what the faucet gave you: keep at least half in ORG, which pays for deployment and all admin transactions.

```
set -a; source backend/.env; set +a
export MST_RPC_URL=https://testnetrpc.mstblockchain.com

for a in $AGENT_ADDRESS $ARBITRATOR_ADDRESS $DEMO_BUYER_ADDRESS $DEMO_SELLER_ADDRESS; do
    cast send $a --value 0.2ether --private-key $ORG_KEY --rpc-url $MST_RPC_URL
done
```

Step 8 — Verify all five balances

```
for r in ORG AGENT ARBITRATOR DEMO_BUYER DEMO_SELLER; do
    a=$(eval echo \${r}_ADDRESS)
    echo "$r $a $(cast balance $a --rpc-url $MST_RPC_URL --ether) MSTC"
done
```

Every wallet must show a non-zero balance. Optionally open each address on the explorer to confirm.

Step 9 — Record and continue

- Tick the funding checkpoint in `docs/STATUS.md` and note the balances (no keys) in `docs/progress/B1.md`.
- Confirm the explorer URL works and add it to `deployments.md`.
- Tell the B1 agent: "wallets funded, continue with deployment (Part 3.4)".

How much MSTC is needed?

Wallet	Pays for	Suggested minimum
ORG	Contract deployment, setKyc, mint, fundFor, gas drips, timeouts	Largest share ($\geq 50\%$ of faucet claim)
AGENT	proposeResolution	0.1-0.2 MSTC
ARBITRATOR	arbitrate	0.1 MSTC
DEMO_BUYER	proposeDeal, release, raiseDispute, acceptResolution	0.2 MSTC
DEMO_SELLER	acceptDeal, markDelivered, acceptResolution	0.2 MSTC

MST fees are very low, so these amounts cover many demo runs. The faucet's exact claim amount and cooldown are not published; adjust accordingly.

Troubleshooting

Faucet page is blank	Enable JavaScript or try another browser; disable aggressive ad/script blockers.
Faucet refuses a second claim	Per-address or per-IP limit. Claim once into ORG and distribute (Step 7), or wait for the cooldown.
Balance stays 0	Check you pasted the ORG address (not another wallet) and are querying chain 91562037. Retry after a minute.
Faucet gives too little / is down	Ask the MST mentors at the venue for testnet MSTC; they typically fund hackathon teams.
cast send: insufficient funds	ORG has less than amount + gas. Send a smaller amount.
Explorer URL does not load	It is community-reported. Use cast balance / cast tx for verification instead.
Key accidentally shown in chat or terminal log	Treat it as leaked: create a new wallet, move the funds, and replace the key in .env.

Final checklist

- ☐ Foundry installed; cast --version works
- ☐ backend/.env is git-ignored
- ☐ 5 wallets created; addresses in deployments.md, keys only in .env
- ☐ ORG funded from the faucet
- ☐ AGENT, ARBITRATOR, DEMO_BUYER, DEMO_SELLER funded from ORG
- ☐ All 5 balances verified non-zero
- ☐ STATUS.md and progress log updated; B1 agent told to deploy

Sources: faucet and testnet details from mstblockchain.com (developer/ecosystem pages and blog); explorer URL from a community MST project and not confirmed by an official MST page.